Insurance Purchase Prediction

This project predicts whether a customer will purchase insurance using machine learning.

Project Overview:

- Data preprocessing and EDA performed
- Models used: Logistic Regression, Random Forest, XGBoost
- Threshold tuning applied for better performance

Dataset:

Anonymized insurance dataset with customer and vehicle features.

Tech Stack:

Python, pandas, numpy, scikit-learn, xgboost, matplotlib, seaborn

Approach:

1. Data preprocessing
2. EDA
3. Model building
4. Evaluation using ROC-AUC, Precision, Recall

Results:

- Best model: XGBoost
- ROC-AUC: ~0.62
- Improved recall with threshold tuning

Future Work:

- Hyperparameter tuning- Deployment using Streamlit/Flask

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